EVERY FORMAL ALGEBRAIC SPACE CARRIES A STRUCTURE SHEAF OF TOPOLOGICAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays three exact sequences of sections, two of them inside the limit computation; all lie inside the counted span and are reproduced verbatim.

Editorial note. The last paragraph compares two fundamental systems of open submodules, one on the sections over the formal space and one on the product over the members of the covering; its final clause is reproduced exactly as delivered, without any rewording or correction.

Editorial note. The counted spans carry 8 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The small etale site of a formal algebraic space and the identification with affine formal algebraic spaces etale over it. Same chapter, Equation equation-etale-topos at proof lines 7669-7671 and Lemma lemma-identify-affine-etale at proof lines 7674-7677. Reduces the construction to a site whose objects are concrete, and identifies reduction as an equivalence with that site.
- Affine formal algebraic spaces as filtered colimits of their defining affine schemes. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 7694-7695. Supplies the filtered system over which the exact sequences are taken.
- Permanence of representability and affineness of a representable morphism. Same chapter, Lemma lemma-permanence-representable at proof lines 7686-7688 and Lemma lemma-affine-representable-by-algebraic-spaces at proof lines 7688-7689. Shows that the members of a covering are representable and hence affine, so that their fibre products are affine schemes.
- Etale morphisms are preserved by the passage considered, and the structure sheaf satisfies the sheaf condition for the etale topology. Properties of Spaces, Lemma spaces-properties-lemma-etale-permanence at proof lines

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7689-7692 and Etale Cohomology, Proposition etale-cohomology-proposition-quasi-coherent-sheaf-fpqc at proof lines 7716-7720. Supplies the exact sequences whose filtered limit gives the sheaf condition at the level of the formal space.

Source reference. Original section 046E: sources/stacks-chow/tags.txt, source lines 5469–5469. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original equation 0DEB: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7413–7413. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03OG: sources/stacks-chow/tags.txt, source lines 4821–4821. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0DEH: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7614–7614. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AQ0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4076–4076. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03FV: sources/stacks-chow/tags.txt, source lines 4506–4506. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda \in \Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Every formal algebraic space has a structure sheaf.*

Proof. Let S be a scheme. Let X be a formal algebraic space over S. By (0DEB) it suffices to construct $\mathcal{O}_X$ as a sheaf of topological rings on $X_{affine,étale}$. Denote $\mathcal{C}$ the category whose objects are morphisms $\varphi : U \to X$ of formal algebraic spaces such that U is an affine formal algebraic space and φ is representable by algebraic spaces and étale. By Lemma 0DEH the functor $U \mapsto U_{red}$ is an equivalence of categories $\mathcal{C} \to X_{affine,étale}$. Hence by the rule given above the lemma, we already have $\mathcal{O}_X$ as a presheaf of topological rings on $X_{affine,étale}$. Thus it suffices to check the sheaf condition.

By definition of $X_{affine,étale}$ a covering corresponds to a finite family $\{g_i : U_i \to U\}_{i=1,\ldots,n}$ of morphisms of $\mathcal{C}$ such that $\{U_{i,red} \to U_{red}\}$ is an étale covering. The morphisms g_i are representably by algebraic spaces (Lemma 0AQ0) hence affine (Lemma 0AKN). Then g_i is étale (follows formally from Properties of Spaces, Lemma 03FV as U_i and U are étale over X in the sense of Bootstrap, Section 046E). Finally, write $U = \operatorname{colim} U_\lambda$ as in Definition 0.1.

With these preparations out of the way, we can prove the sheaf property as follows. For each λ we set $U_{i,\lambda} = U_i \times_U U_\lambda$ and $U_{ij,\lambda} = (U_i \times_U U_j) \times_U U_\lambda$. By the above, these are affine schemes, $\{U_{i,\lambda} \to U_\lambda\}$ is an étale covering, and $U_{ij,\lambda} = U_{i,\lambda} \times_{U_\lambda} U_{j,\lambda}$. Also we have $U_i = \operatorname{colim} U_{i,\lambda}$ and $U_i \times_U U_j = \operatorname{colim} U_{ij,\lambda}$. For each λ we have an exact sequence

$$0 \to \Gamma(U_\lambda, \mathcal{O}_{U_\lambda}) \to \prod\nolimits_i \Gamma(U_{i,\lambda}, \mathcal{O}_{U_{i,\lambda}}) \to \prod\nolimits_{i,j} \Gamma(U_{ij,\lambda}, \mathcal{O}_{U_{ij,\lambda}})$$

as we have the sheaf condition for the structure sheaf on U_λ and the étale topology (see Étale Cohomology, Proposition 03OG). Since limits commute with limits, the inverse limit of these exact sequences is an exact sequence

$$0 \to \lim \Gamma(U_\lambda, \mathcal{O}_{U_\lambda}) \to \prod\nolimits_i \lim \Gamma(U_{i,\lambda}, \mathcal{O}_{U_{i,\lambda}}) \to \prod\nolimits_{i,j} \lim \Gamma(U_{ij,\lambda}, \mathcal{O}_{U_{ij,\lambda}})$$

This means that

$$0 \to \mathcal{O}_X(U_{red}) \to \prod\nolimits_i \mathcal{O}_X(U_{i,red}) \to \prod\nolimits_{i,j} \mathcal{O}_X((U_i \times_U U_j)_{red})$$

is exact as a sequence of abelian groups. We still have to show that the limit topology on $\mathcal{O}_X(U_{red})$ is induced from the limit topology on $\prod_{i=1,\ldots,n} \mathcal{O}_X(U_{i,red})$. The kernels of the maps $\mathcal{O}_X(U_{red}) \to \mathcal{O}_{U_\lambda}(U_\lambda)$ form a fundamental system of open submodules. Since these are the inverse images of the kernels of the maps $\prod_i \mathcal{O}_X(U_{i,red}) \to \prod_i \mathcal{O}_{U_{i,\lambda}}(U_{i,\lambda})$ and these kernels form a fundamental system of open submodules of $\mathcal{O}_X(U_{red}) \to \mathcal{O}_{U_\lambda}(U_\lambda)$ we win. □